WORKED SOLUTION

STEP II 1996, Question 5

Pure • Roots of Unity

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q5

The key idea

The quartic is *palindromic*: its coefficients read the same forwards and backwards. Dividing by z^2 groups the terms into $z^2 + z^{-2}$ and $z + z^{-1}$, and both are expressible in terms of $u = z + z^{-1}$, so a quartic in z collapses to a quadratic in u. The second half recognises the quartic as the factor left over when $z - 1$ is taken out of $z^5 - 1$, so its roots are fifth roots of unity; comparing the two descriptions of the same four numbers gives exact values of trigonometric functions.

Solution

The values of u

Since $z = 0$ does not satisfy $(*)$, we may divide by z^2:

$$z^2 + z + 1 + z^{-1} + z^{-2} = 0.$$

Now $u^2 = (z + z^{-1})^2 = z^2 + 2 + z^{-2}$, so $z^2 + z^{-2} = u^2 - 2$. The equation becomes

$$u^2 - 2 + u + 1 = 0, \qquad \text{that is} \qquad u^2 + u - 1 = 0,$$

so

$$u = \frac{-1 + \sqrt{5}}{2} \quad \text{or} \quad u = \frac{-1 - \sqrt{5}}{2}.$$

The values of z

From $z + z^{-1} = u$ we get $z^2 - uz + 1 = 0$, so

$$z = \frac{u \pm \sqrt{u^2 - 4}}{2}.$$

Using $u^2 = 1 - u$, the discriminant is $u^2 - 4 = -3 - u$, which is negative for both values of u (as $|u| < 2$ in each case), so the roots are non-real:

$$z = \frac{u \pm i\sqrt{3+u}}{2}.$$

For $u = \frac{-1+\sqrt{5}}{2}$ we have $3 + u = \frac{5+\sqrt{5}}{2}$, and for $u = \frac{-1-\sqrt{5}}{2}$ we have $3 + u = \frac{5-\sqrt{5}}{2}$. Writing $\sqrt{\frac{5\pm\sqrt{5}}{2}} = \frac{1}{2}\sqrt{10 \pm 2\sqrt{5}}$, the four solutions of $(*)$ are

$$z = \frac{-1 + \sqrt{5} \pm i\sqrt{10 + 2\sqrt{5}}}{4} \qquad \text{and} \qquad z = \frac{-1 - \sqrt{5} \pm i\sqrt{10 - 2\sqrt{5}}}{4}.$$

Each value of u gives two values of z (a pair z, z^{-1}, which here are complex conjugates), and a quartic has exactly four roots, so these are all of them.

The link with $z^5 = 1$

Multiplying out,

$$(z-1)(z^4 + z^3 + z^2 + z + 1) = z^5 - 1,$$

since the middle terms telescope. So if z satisfies $(*)$, the second factor is zero and $z^5 - 1 = 0$.

The solutions of $z^5 = 1$ are $z = \cos\frac{2k\pi}{5} + i\sin\frac{2k\pi}{5}$ for $k = 0, 1, 2, 3, 4$. The value $k = 0$ gives $z = 1$, which does *not* satisfy $(*)$ (the left-hand side is 5). The other four are distinct and all satisfy $(*)$, because for them $z - 1 \neq 0$ and so the factor $z^4 + z^3 + z^2 + z + 1$ must vanish. Hence the solutions of $(*)$ are

$$z = \cos\theta + i\sin\theta, \qquad r = 1, \quad \theta = \pm\frac{2\pi}{5}, \pm\frac{4\pi}{5}.$$

The deduction

We now have two descriptions of the same four numbers, and we must match them up.

? **Ask yourself** Which of the four surd expressions is $\cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}$?

Look at signs. Since $0 < \frac{2\pi}{5} < \frac{\pi}{2}$, the number $\cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}$ has positive real part and positive imaginary part. Among the surd expressions, the real parts are $\frac{-1+\sqrt{5}}{4} > 0$ and $\frac{-1-\sqrt{5}}{4} < 0$, so exactly one of them lies in the first quadrant:

$$\cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5} = \frac{-1+\sqrt{5}}{4} + i\,\frac{\sqrt{10+2\sqrt{5}}}{4}.$$

(The other angle in $(0, \pi)$, namely $\frac{4\pi}{5}$, has negative cosine, so it is matched with the other root with positive imaginary part.) Equating real and imaginary parts,

$$\sin\frac{2\pi}{5} = \frac{\sqrt{10+2\sqrt{5}}}{4} \qquad \text{and} \qquad \cos\frac{2\pi}{5} = \frac{-1+\sqrt{5}}{4},$$

as required. As a by-product, $\cos\frac{4\pi}{5} = \frac{-1-\sqrt{5}}{4}$ and $\sin\frac{4\pi}{5} = \frac{\sqrt{10-2\sqrt{5}}}{4}$.

Checks

- $\cos\frac{2\pi}{5} = \cos 72^\circ \approx 0.30902$ and $\frac{-1+\sqrt{5}}{4} \approx 0.30902$; $\sin 72^\circ \approx 0.95106$ and $\frac{1}{4}\sqrt{10+2\sqrt{5}} \approx 0.95106$.
- The sum of the four roots should be -1 (minus the coefficient of z^3): the real parts add to $\frac{2(-1+\sqrt{5})+2(-1-\sqrt{5})}{4} = -1$. Equivalently, the two values of u add to -1.
- Each root should have modulus 1: for example $\frac{(-1+\sqrt{5})^2+(10+2\sqrt{5})}{16} = \frac{16}{16} = 1$.

We checked numerically that all four surd expressions satisfy $(*)$ and $z^5 = 1$ and coincide with $e^{\pm 2\pi i/5}$, $e^{\pm 4\pi i/5}$.

Common slips

- Forgetting to exclude $z = 1$: it satisfies $z^5 = 1$ but not $(*)$, so $(*)$ has four roots, not five.
- Matching the surds to the angles without a reason. The sign of the real part is what pins down which root is $\cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}$.
- Writing $z^2 + z^{-2} = u^2$ and losing the -2.

Beyond STEP

The substitution $u = z + z^{-1}$ halves the degree of any palindromic polynomial. The fact that $\cos\frac{2\pi}{5}$ is reached by solving quadratics only is no accident: Gauss showed that $\cos\frac{2\pi}{n}$ can be written with square roots alone exactly when a regular n-gon can be constructed with ruler and compass, and this question is the algebra behind constructing the regular pentagon. The full story is Galois theory, met in a second-year algebra course; Needham's *Visual Complex Analysis* gives a beautiful geometric view of roots of unity.

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